Frontiers of Science

FINAL EXAM STUDY GUIDE, FALL 2025
Scientific Habits of Mind Learning Objectives

STATISTICS (Smart Sparrow lessons 1&3, Tutorial)

1. Demonstrate an understanding of mean, standard deviation, and standard error.
2. Estimate mean and standard deviation from a distribution (histogram).
3. Calculate/estimate ~95% confidence intervals (mean ± 2SE intervals).
4. Determine statistical significance from confidence intervals and p-values (for bar graphs and correlations/scatter plots).
5. Distinguish between random and systematic errors and describe their respective implications for precision and accuracy.

CALCULATIONS (Smart Sparrow lessons 2&5, Tutorials)

6. Use units and unit conversions in calculations, utilize dimensional analysis.
7. Manipulate algebraic expressions and powers of ten.
8. Identify when to use back-of-the-envelope calculations, make estimates using back-of-the-envelope calculations, identify and state assumptions. Check answer plausibility (does your answer make sense?), put your answer into context by taking ratios/determining order of magnitude difference.

SCIENTIFIC CLAIMS AND PROCESS (Smart Sparrow lesson 4, Tutorial)

9. Describe the role of falsifiability in science.
10. Distinguish between scientific hypotheses and scientific theories.
11. Consider experimental study design, including use of proxies; determine the purpose of controls in an experiment, assess what makes a good control group or control task.
12. Distinguish between observations and experiments, identify strengths and weaknesses of experimental and observational studies.
13. Distinguish between correlation and causation. Identify potential explanations for an observed correlation and evaluate whether a causal relationship is well supported.
14. Evaluate hypotheses in the context of presented evidence, propose experiments to test hypotheses or extend previous studies.
15. Read and interpret graphical information in histograms, bar graphs, scatter plots, and time series using axes labels, numerical information and units, error bars, and figure captions.
16. Consider the role of models in understanding physical processes and future climate, utilize models, and identify their limitations.

SENSE OF SCALE (Smart Sparrow lessons 2&6, Tutorial)

17. Demonstrate a sense of scale for the structure and function of molecules of life, issues relevant to the current pandemic (e.g. infection rates, population sizes), and issues relevant to global warming (time scales, modern increases in temperature and greenhouse gases vs. past trends, predictions from future models). Use benchmarks and analogies to put quantities into context.
18. Utilize logarithmic scales, read values off a log scale, add values to a log scale. Identify when a log scale is useful and when it cannot be used.

PROBABILITY (Smart Sparrow lessons 7&8, Tutorial)

19. Assign probabilities to events, estimate probabilities of events.
20. Calculate probabilities of multiple events (independent events, mutually exclusive events, both, and/or).
21. Calculate probabilities of events occurring at least once, given N opportunities. Approximate probabilities of rare events.

FEEDBACKS (Smart Sparrow lesson 9, Tutorial)

24. Identify feedback loops and draw feedback loops.
25. Determine whether a feedback loop is positive or negative.

Concept Learning Objectives

Molecules and Us Week 1: Chemistry of Life and Molecular Machines

1. Develop a working vocabulary of key terms in (1) Biophysical Chemistry: chemical bonds (covalent, ionic, and hydrogen bonds), macromolecules, biomolecules, monomers (amino acids and nucleotides), biopolymers (DNA, RNA, and proteins), protein structure (primary; secondary: alpha helix and beta sheet; and tertiary), and nucleic acid structure (primary; secondary: double helix; and tertiary); AND (2) Genetics: genes, chromosomes, and genomes.

2. List the major biomolecules involved in each of the following molecular processes in the cell: replication, transcription, and translation. Discuss the roles of these processes and biomolecules in the context of the Central Dogma of Molecular Biology.

3. Explain how light and other forms of radiation interact with biomolecules and how those interactions can be used to study the structural and dynamic features of biomolecules. List two recent techniques used to study biopolymers that are based on these interactions.

4. Describe how scientists use computational tools, such as PyMOL, to access publicly available databases in efforts to understand diseases and how to combat them.

Molecules and Life Week 2: Molecular Tools, Diseases, and Therapeutics

1. Describe the molecular basis for sickle cell anemia.

2. Summarize the goals of genome engineering and how these can be achieved using the CRISPR Cas9 system. Assess ethical arguments pro and con the use of such genome engineering in humans.

3. List leading causes of death and leading causes of cancer deaths in the United States and world-wide, and discuss where disparities may be seen and why.

4. Discuss the general process scientists use to identify and develop therapeutics for different diseases.

5. Outline the steps involved in cancer development and explain which ones can be targeted to develop therapeutics, using two specific examples.

Climate and Us Week 1: Global Warming: What We Know and What We Don’t Know

1. Distinguish between weather and climate.

2. Explain the role that astronomical influences (sunspots and eccentricity, obliquity, and precession) play in climate.

3. Analyze the composition of the Earth's atmosphere, the recent changes that we observe, the role humans play in those changes, and how we know human activity is the cause.

4. Describe the global temperature record since 1880, with a focus on the last few decades.

5. Reconstruct the Earth's energy balance by considering the flow of energy into and away from the Earth's surface and atmosphere, with a special emphasis on the role of reflectivity and greenhouse gases.

Climate and Us Week 2: Past, Present, and Future Climate

1. Describe what paleoclimate data (from tree rings and ice cores) tell us about the long-term history of Earth's climate. Compare the magnitude of the current warming and greenhouse gas concentration changes to past trends.

2. Describe the process scientists use to predict future climate and its impacts. Identify the importance and limitations of climate modeling, including the role of feedbacks in the climate system.

3. Analyze fictions used in arguments for and against taking action to mitigate global warming.

4. List promise and problems of approaches we can use to address current and future challenges brought about by global warming.